

Basis Data Part9

PENGAMPU:

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Using OpenRefine



Now let's practice cleaning some data. Download [sample-file.csv](#)

Loading a file and creating a project



A power tool for working with messy data.

Create Project

Open Project

Import Project

Language Settings

Create a project by importing data. What kinds of data files can I import?

TSV, CSV, *SV, Excel (.xls and .xlsx), JSON, XML, RDF as XML, and Google Data documents are all supported

Get data from

This Computer

Web Addresses (URLs)

Clipboard

Google Data

Locate one or more files on your computer to upload:

Choose Files No file chosen

Next »

Loading a file and creating a project

Refine OPEN A power tool for working with messy data.

Create Project Open Project Import Project Language Settings

« Start Over Configure Parsing Options Project name Open Refine Data Cleaning Exercise Create Project »

	Timestamp	Name of person presenting	Name of project/news package
1.	4/20/17 15:53	Tim Campbell	Life on Hold
2.	4/20/17 15:53	Timmy Campbell	Life on hold
3.	4/20/17 15:53	Timothy Campbell	Life on Hold
4.	4/20/17 15:53	Timothy Campbell	Life on Hold - Syrian Refugees in Lebanon
5.	4/20/17 15:53	Tim	Life On Hold
6.	4/20/17 15:53	Campbell	Life on Hold
7.	4/20/17 15:53	Tim Campbell	Life on Hold: Stories of Syrian Refugees in Lebanon
8.	4/20/17 15:53	Timmy	Life on Hold
9.	4/20/17 15:53	Tim Campbell	Life on Hold
10.	4/20/17 15:54	Timothy Campbell	Life on Hold
11.	4/20/17 15:54	Timothy Campbell	Life on Hold
12.	4/20/17 15:54	Tim Campbell	Life on Hold
13.	4/20/17 15:54	Tim	Life on Hold
14.	4/20/17 15:55	Tim	Life On Hold
15.	4/20/17 15:55	Tim Campbell	Life on Hold
16.	4/20/17 15:57	Timmy Campbell	Life In Hold Stories of Syrian refugees in Lebanon
17.	4/20/17 16:01	Theresa Coats	Look at This
18.	4/20/17 16:04	Teresa Coats	NPR's Look at This
19.	4/20/17 16:04	Teresa Coats	Look At This

Parse data as

CSV / TSV / separator-based files

Line-based text files

Fixed-width field text files

PC-Axis text files

JSON files

MARC files

RDF/N3 files

XML files

Open Document Format spreadsheets (.ods)

Character encoding

Update Preview

Columns are separated by

☒ commas (CSV)

☐ tabs (TSV)

☐ custom ,

Escape special characters with \

☐ Ignore first 0 line(s) at beginning of file

☒ Parse next 1 line(s) as column headers

☐ Discard initial 0 row(s) of data

☐ Load at most 0 row(s) of data

☐ Parse cell text into numbers, dates, ...

☒ Quotation marks are used to enclose cells containing column separators

☒ Store blank rows

☒ Store blank cells as nulls

☐ Store file source (file names, URLs) in each row

Version 2.7 [TRUNK]

Help About

Examining our data

Click the arrow on the “**Name of Person**” column, and select “**Facet, “Text Facet.”**”

You’ll see a window pop up on the left hand side of the screen. This gives us an overview of the values in that column – which, in this case, is student names.

You’ll notice that a lot of data has been entered **inconsistently**. **Alex Castillo**, for example, is entered as **Alexander, Alexander Castillo, Alex Castilloooooo**. This inconsistency makes things tricky later down the line when you’re trying to analyze your data because your computer will treat Alex Castillo and Alex Castillooooooo as **different people**, even though we as humans know they’re the **same person**.

Removing whitespace

Alex Castillo 9
Alex Castillo 1

The reason we're seeing two entries is because one entry **has a space** following it. Again, our computer reads this as two separate people, even though we as humans know better.

Click the small arrow next to the **"Name of person"** column. In the menu, select **"Edit Cells,"** **"Common Transformations,"** **"Trim leading and trailing whitespace."**

Changing the case

evelyn wong 1

Evelyn Wong 14

Click the small arrow next to the **“Name of person”** column. In the menu, select **“Edit Cells,”** **“Common Transformations,”** **“To Titlecase.”**

▼ Name of person	▼ Name of project/news package	
Facet	▶	on Hold
Text filter	▶	on hold
▶ Edit cells	▶ Transform...	anon
▶ Edit column	▶ Common transforms	▶ Trim leading and trailing whitespace
▶ Transpose	▶ Fill down	▶ Collapse consecutive whitespace
Sort...	▶ Blank down	▶ Unescape HTML entities
▶ View	▶ Split multi-valued cells...	▶ To titlecase

Cleaning through cluster and edit

Show as: rows records		Show: 5 10 25 50 rows	
▼ All	▼ Timestamp	▼ Name of person	▼ Name of project/news package
☆	🗨	1.	4/20/17 15:53
☆	🗨	2.	4/20/17 15:53
☆	🗨	3.	4/20/17 15:53
☆	🗨	4.	4/20/17 15:53
☆	🗨	5.	4/20/17 15:53
☆	🗨	6.	4/20/17 15:53
☆	🗨	7.	4/20/17 15:53
☆	🗨	8.	4/20/17 15:53
☆	🗨	9.	4/20/17 15:53
☆	🗨	10.	4/20/17 15:54

Facet

Text filter

Edit cells

Edit column

Transpose

Sort...

View

Reconcile

Keep on Hold

Keep on hold

Keep on Hold

Transform...

Common transforms

Fill down

Blank down

Split multi-valued cells...

Join multi-valued cells...

Cluster and edit...

Cleaning through cluster and edit

Cluster & Edit column "Name of person presenting"

This feature helps you find groups of different cell values that might be alternative representations of the same thing. For example, the two strings "New York" and "new york" are very likely to refer to the same concept and just have capitalization differences, and "Gödel" and "Godel" probably refer to the same person. [Find out more ...](#)

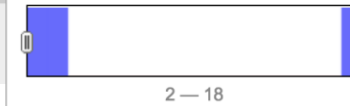
Method key collision

Keying Function fingerprint

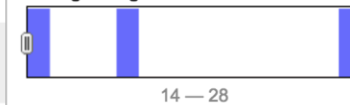
3 clusters found

Cluster Size	Row Count	Values in Cluster	Merge?	New Cell Value
2	3	- Sheila Rhodes & Jake Wheeler (2 rows) - Sheila Rhodes, Jake Wheeler (1 rows) Browse this cluster	☐	Sheila Rhodes & Jake Wheeler
2	18	- Candice Washington (17 rows) - Washington Candice (1 rows)	☐	Candice Washington
2	2	- Jay And Sheila (1 rows) - Sheila And Jay (1 rows)	☐	Jay And Sheila

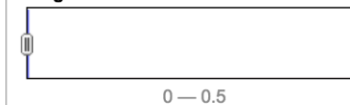
Rows in Cluster



Average Length of Choices



Length Variance of Choices



To clean any given name, all we have to do is check the box under the **Merge?** column and click the **Merge Selected & Recluster** button.

Once we do, the variations of the name in the Values in Cluster Column will merge under the new name we've chosen in the New Cell Value column.

Select All Unselect All

Export Clusters

Merge Selected & Re-Cluster

Merge Selected & Close

Close

Algorithm settings

Keying Function

✓ fingerprint

nggram-fingerprint

metaphone3

cologne-phonetic

cted method

Cluster Size	Row Count	Values in Cluster	Merge?	New Cell Value
2	3	- Hattie Garrett (2 rows) - Hatie Garrett (1 rows)	☐	Hattie Garrett
2	7	- Sheila Rhodes, Jacob Wheeler (6 rows) - Sheila Rhodes/jacob Wheeler (1 rows)	☐	Sheila Rhodes, Jacob Wheeler
2	13	- Hattie Garrett (10 rows) - Hattie Garret (3 rows)	☐	Hattie Garrett

Cleaning individual entries

✕ **Name of person presenting** change

59 choices Sort by: **name** count Cluster

Alexander	1
Alexander Castillo	15
Andrew V.	1
Andrew Vasquez	12
Andy Vasquez	7

We can clean those up manually by simply clicking edit next to the name in the text facet window and renaming the names we want to change.

Exporting cleaned data

When you're finished, you can export your cleaned dataset as a CSV by clicking "Export" at the top of your screen and selecting "Comma Separated Value."

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